

Jin Yang, Ph.D.

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Education

- Sep. 2019 – Aug. 2026 📖 **Combined M.S. – Ph.D. in Computer Science and Engineering**
Institution: Ulsan National Institute of Science and Technology (UNIST), Korea.
Thesis: *Exploiting Data and Workload Characteristics for Efficient Data Management Systems*.
Advisor: Prof. Young-ri Choi (System Software Laboratory).
- Mar. 2015 – Aug. 2019 📖 **B.S. in Computer Science and Engineering (major) and Mathematics (minor)**
Institution: Ulsan National Institute of Science and Technology (UNIST), Korea.

Research interests

- Storage Systems 📖 Indexing and caching for key-value stores
📖 Designs that adapt to the data distribution and the workload
📖 Data management in storage engines

Research Publications

Conference Proceedings

- 1 **Jin Yang**, Hyeonseok Oh, and Young-ri Choi, “TreePass: Lightweight Tree-Guided Cache Management for B+-tree Key-Value Stores,” in *Proceedings of the 25th USENIX Conference on File and Storage Technologies (FAST ’27)*, 2027, Acceptance rate: 19.6% (18/92, Spring cycle).
- 2 Heejin Yoon, **Jin Yang**, Juyoung Bang, Sam H. Noh, and Young-ri Choi, “Advocating for Key-Value Stores with Workload Pattern Aware Dynamic Compaction,” in *Proceedings of the 16th ACM Workshop on Hot Topics in Storage and File Systems (HotStorage ’24)*, 2024. 📄 DOI: 10.1145/3655038.3665955.
- 3 **Jin Yang**, Heejin Yoon, Gyeongchan Yun, Sam H. Noh, and Young-ri Choi, “DyTIS: A Dynamic Dataset Targeted Index Structure Simultaneously Efficient for Search, Insert, and Scan,” in *Proceedings of the 18th European Conference on Computer Systems (EuroSys ’23)*, 2023, pp. 800–816. 📄 DOI: 10.1145/3552326.3587434, Acceptance rate: 16.7% (54/323).

Journal Articles

- 1 **Jin Yang**, Heejin Yoon, Gyeongchan Yun, Sam H. Noh, and Young-ri Choi, “A Dynamic Characteristic Aware Index Structure Optimized for Real-world Datasets,” *ACM Transactions on Storage*, vol. 21, no. 2, pp. 1–32, 2025. 📄 DOI: 10.1145/3707642.

Patents

- 1 Young-ri Choi, **Jin Yang**, Heejin Yoon, Gyeongchan Yun, and Sam H. Noh, *Method and Apparatus for Performing Dynamic Hashing to Store Key Values in an Ordered State*, KR Patent 10-2683404 (granted Jul. 2024); US Patent App. Pub. US 2024/0048358 A1 (Feb. 2024), 2024.
- 2 Young-ri Choi, Songyi Lee, **Jin Yang**, and Sam H. Noh, *Method and Apparatus for Converting Data Key Value*, KR Patent 10-2388458 (granted Apr. 2022), 2022.

Awards and honors

- 2023  **Invited Speaker, Top Conference Session**, KCC 2023, Jeju, Korea — DyTIS (EuroSys '23)
-  **EuroSys 2023 Grant**, EuroSys Conference
- 2018  **UNIST Hackathon Alkkago Award**, sponsored by Naver D2SF (500,000 KRW)
- 2016  **UNIST HeXATHON Finalist (Top 12 teams)**, sponsored by Naver D2 and UNIST ECE

Software Artifacts

- TreePass**  An open-source repository for "TreePass: Lightweight Tree-Guided Cache Management for B+-tree Key-Value Stores" accepted at FAST '27.
(<https://github.com/unist-ssl/TreePass>)
- DyTIS**  An open-source repository for "DyTIS: A Dynamic Dataset Targeted Index Structure Simultaneously Efficient for Search, Insert, and Scan" accepted at EuroSys '23.
(<https://github.com/unist-ssl/DyTIS>)
- HASS**  A repository for Hotness-Aware Sharding Split for MongoDB.
(<https://github.com/heejin5178/HASS>)
- RH-Skiplist**  A repository for an indexing structure combining a skiplist and a hash for hybrid memory systems with Intel Optane persistent memory and DRAM.
(<https://github.com/sheepjin11/RH-Skiplist>)

Teaching Experience

- Mar. 2022 – Aug. 2022  Teaching assistant, Introduction to Human Computer Interaction, undergraduate course at UNIST.
- Sep. 2021 – Feb. 2022  Teaching assistant, Data Structures, undergraduate course at UNIST.
- Mar. 2021 – Aug. 2021  Teaching assistant, Advanced Programming, undergraduate course at UNIST.
- Sep. 2020 – Feb. 2021  Teaching assistant, Operating Systems, undergraduate course at UNIST.
- Mar. 2020 – Aug. 2020  Teaching assistant, Computer Networks, undergraduate course at UNIST.
- Sep. 2019 – Feb. 2020  Teaching assistant, Introduction to AI Programming 2, undergraduate course at UNIST.
- Sep. 2016 – Feb. 2017  Teaching assistant, Linear Algebra, undergraduate course at UNIST.

Professional Experience

- Mar. 2017 – Dec. 2019  Member (5th) and Organizer (6th, 7th), Like Lion, UNIST — a student-run programming and web development community.
- Dec. 2017 – Feb. 2018  Intern developer, DoDotDo.